

On (H6): the local second moment of the interaction density
what the gap cannot do, and how much less than (H6) suffices

Abstract

(H6) asks for $\sup_g \|V_B \Omega_g\| < \infty$, V_B a local Wick polynomial. **I do not have a proof.** This note contributes three things.

(1) A negative result: the gap is the wrong tool, for an ultraviolet reason.

The two natural gap-based routes are *vacuous*. Both pass through quantities that are UV divergent while $\|V_B \Omega_g\|$ itself is finite. The f -sum rule gives $\langle V_B \Omega_g, H(g) V_B \Omega_g \rangle = \frac{1}{2} \omega_g([V_B, [H_0, V_B]])$ — note the double commutator is g -independent, because $[V(g), V_B] = 0$ — and equals $\frac{1}{2} \omega_g(\int h^2(:P'(\varphi):)^2)$, a Wick square at coincident points. In the free case it diverges logarithmically; verified numerically (§2).

(2) A positive result: (H6) can be weakened to (H6'), uniform $L^{1+\delta}$. Truncating V_B at height M and using that the *exact light cone* makes the truncation tail *commute* with $\alpha_t(A)$ for $|t| < R$, one needs only a first moment of the tail. This costs a factor $\delta/(1+\delta)$ in the clustering rate (Theorem 3.2). A first moment alone — which *is* available, uniformly, from Guerra–Rosen–Simon — is not enough; uniform integrability is exactly what is missing.

(3) A sharp identification of what remains. Via Hölder and Nelson hypercontractivity, (H6') follows from a uniform local L^q bound on the ground-state density for *some* $q > 1$, i.e. from Problem 1 of [7] verbatim; and via localised N_τ estimates, (H6) follows from the deg P -th moment of the local number operator, strictly stronger than (H2)-full's first moment (§4).

1 Setting; (H6) in Q -space

Notation as in [1, 2, 3]. Let $h \in C_c^\infty(B)$, $|h| \leq 1$, and

$$V_B := \int_B h(x) :P(\varphi_0(x)): dx,$$

a symmetric operator affiliated to $\mathfrak{A}(B)$. In the Q -space representation V_B is a *multiplication* operator and $d\mu_g := \Omega_g^2 d\varphi_0$ is a probability measure, so

$$\|V_B \Omega_g\|^2 = \mathbb{E}_{\mu_g}[V_B^2], \tag{1}$$

and (H6) reads: *the second moment of a fixed local random variable is bounded uniformly in the cutoff*. For each fixed g it is finite, since $\Omega_g \in L^p(Q, d\varphi_0)$ for all $p < \infty$ and $V_B \in L^p$ for all $p < \infty$ by Nelson's hypercontractive bound; only the uniformity in g is at stake.

Two facts are available *uniformly* in g and will be used repeatedly:

(F₁) Uniform first moment. By [6, Thm. 2], $\pm V_B \leq \varepsilon H(g) + c(B)$ with c independent of g ; taking ω_g and using $H(g)\Omega_g = 0$, $|\omega_g(V_B)| \leq c(B)$, and moreover $\|(H(g) + c)^{-1/2} V_B (H(g) + c)^{-1/2}\| \leq 1 + \varepsilon$.

(F₂) Uniform local excitation energy. $\|H(g)^{1/2} A \Omega_g\|^2 = -i \omega_g(A^* \delta(A))$ by [1, Lem. 2.1], bounded uniformly in g for A in the class covered by the φ -bound of [5].

2 The gap cannot prove (H6)

Proposition 2.1 (*f*-sum rule; the double commutator is *g*-independent). *For symmetric X with $X\Omega_g \in D(H(g)^{1/2})$,*

$$\langle X\Omega_g, H(g)X\Omega_g \rangle = \frac{1}{2} \omega_g([X, [H(g), X]]).$$

If $X = V_B$ then, because V_B and the interaction $V(g)$ are both functions of φ_0 alone and hence commute,

$$[V_B, [H(g), V_B]] = [V_B, [H_0, V_B]] = \int h(y)^2 (:P'(\varphi_0(y)):)^2 dy.$$

Proof. $H(g)\Omega_g = 0$ gives $\omega_g([X, [H, X]]) = 2\langle X\Omega_g, HX\Omega_g \rangle$ directly. For the second claim, $[V(g), V_B] = 0$ and $[E(g), V_B] = 0$, so only H_0 contributes; only the $\frac{1}{2} \int : \pi^2 :$ term of H_0 fails to commute with a function of φ_0 . With $[V_B, \pi(y)] = i h(y) :P'(\varphi_0(y)):$ one computes $[H_0, V_B] = -\frac{i}{2} \int h(y) (\pi(y) :P': + :P': \pi(y)) dy$ and then the stated identity. $\square$

Combining Proposition 2.1 with the gap $H(g) \geq \gamma(1 - P_g)$ gives the tempting

$$\|V_B\Omega_g\|^2 \leq \omega_g(V_B)^2 + \frac{1}{2\gamma} \omega_g\left(\int h^2(:P'(\varphi_0):)^2\right), \quad (2)$$

whose first term is uniformly bounded by (F₁) and whose second term is *g*-independent. This looks like a proof of (H6). It is not.

Proposition 2.2 (the bound is vacuous). *The right-hand side of (2) is $+\infty$. Already for $P(\xi) = \xi^2$ and the Fock vacuum,*

$$\langle V_B\Omega_0, H_0V_B\Omega_0 \rangle = c \iint |\hat{h}(k_1 + k_2)|^2 \frac{\mu(k_1) + \mu(k_2)}{\mu(k_1)\mu(k_2)} dk_1 dk_2 = +\infty,$$

logarithmically divergent, while $\|V_B\Omega_0\|^2 = c \iint |\hat{h}(k_1 + k_2)|^2 (\mu_1\mu_2)^{-1} < \infty$.

Proof. $V_B\Omega_0$ is the two-particle vector with kernel $\propto \hat{h}(k_1 + k_2)(\mu_1\mu_2)^{-1/2}$. Writing $K = k_1 + k_2$, the norm integral is $\int |\hat{h}(K)|^2 \int dk (\mu(k)\mu(K - k))^{-1}$, convergent since the inner integrand is $O(k^{-2})$; the energy integral is $\int |\hat{h}(K)|^2 \int dk (\mu(k)^{-1} + \mu(K - k)^{-1})$, whose inner integrand is $O(|k|^{-1})$. $\square$

Numerical confirmation ($m = 1$, $|\hat{h}(K)|^2 = e^{-K^2/2}$, symmetric cutoff $|k_i| \leq \Lambda$):

Λ	$\ V_B\Omega_0\ ^2$	$\langle V_B\Omega_0, H_0V_B\Omega_0 \rangle$
20	7.2493	36.79
80	7.4419	50.84
320	7.4892	64.78
640	7.4970	71.73

The norm saturates; the energy increases by ≈ 6.96 per doubling of Λ — exactly logarithmic.

Remark 2.3 (the resolvent route fails identically). From $H(g)V_B\Omega_g = [H_0, V_B]\Omega_g$ one gets $\|(1 - P_g)V_B\Omega_g\| \leq \gamma^{-1} \|[H_0, V_B]\Omega_g\|$; but $\|[H_0, V_B]\Omega_0\| = \infty$ by the same computation. Vacuum again.

Remark 2.4 (diagnosis). $\|V_B \Omega_g\|^2$ is *ultraviolet-sensitive*: in Q -space it is $\mathbb{E}_{\mu_g}[V_B^2]$, and the Wick reordering $:P(\varphi(x))::P(\varphi(y)):=\sum_k \frac{C(x-y)^k}{k!} :P^{(k)}(\varphi(x))P^{(k)}(\varphi(y)):$ makes it depend on the short-distance behaviour $C(x-y) \sim -\frac{1}{2\pi} \log|x-y|$. The spectral gap is an *infrared* object. Any argument routed through $H(g)^{-1}(1-P_g)$ or through a double commutator discards the Wick-ordering structure and produces a divergent majorant. **The tools that can work are hypercontractivity, N_τ estimates and local L^p bounds — not the gap.**

3 (H6) can be weakened to (H6')

Hypothesis 3.1 ((H6')). There are $\delta > 0$ and $c_\delta(B) < \infty$ with $\sup_{g \in \mathcal{G}} \mathbb{E}_{\mu_g}[|V_B|^{1+\delta}] \leq c_\delta(B)$.

The point of the next theorem is that the *exact light cone* makes the truncation tail commute with the evolved observable, so that only a *first* moment of the tail is needed where a Cauchy–Schwarz step would have demanded a second.

Theorem 3.2 (truncation). *Let $A \in \mathfrak{A}(O)$, $R := \text{dist}(O, B) > 0$, and let g satisfy $d(g, O) > R$ and $\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty)$. Put $\Psi(t) := \omega_g(V_B \alpha_t(A)) - \omega_g(V_B) \omega_g(A)$. Assume (H6'). Then for all $|t| < R$,*

$$|\Psi(t)| \leq 2(1 + c_\delta(B)) \|A\| e^{-\gamma'(R-|t|)}, \quad \gamma' := \frac{\gamma \delta}{2(1+\delta)}.$$

Proof. Fix $M > 0$ and set $V^{(M)} := V_B \mathbf{1}_{\{|V_B| \leq M\}} \in \mathfrak{A}(B)$, a bounded multiplication operator with $\|V^{(M)}\| \leq M$, and $Y := V_B - V^{(M)} = V_B \mathbf{1}_{\{|V_B| > M\}}$. Write $Y = Y_+ - Y_-$ with $Y_\pm \geq 0$, both affiliated to $\mathfrak{A}(B)$.

Step 1 (the tail costs only a first moment). Let $|t| < R$. By the exact light cone [1, Fact 1.3(i)], $\alpha_t(A) \in \mathfrak{A}(O_{|t|})$ and $O_{|t|} \cap B = \emptyset$, so $\alpha_t(A)$ commutes with $Y_\pm$ and hence with $Y_\pm^{1/2}$. Therefore

$$|\langle Y_\pm \Omega_g, \alpha_t(A) \Omega_g \rangle| = |\langle Y_\pm^{1/2} \Omega_g, \alpha_t(A) Y_\pm^{1/2} \Omega_g \rangle| \leq \|A\| \|Y_\pm^{1/2} \Omega_g\|^2 = \|A\| \mathbb{E}_{\mu_g}[Y_\pm].$$

Adding the two signs and the disconnected term,

$$|\Psi(t) - \Psi^{(M)}(t)| \leq 2\|A\| \mathbb{E}_{\mu_g}[|V_B| \mathbf{1}_{\{|V_B| > M\}}] \leq 2\|A\| c_\delta(B) M^{-\delta},$$

the last step by $|V_B| \mathbf{1}_{\{|V_B| > M\}} \leq |V_B|^{1+\delta} M^{-\delta}$ and (H6'). Here $\Psi^{(M)}$ is Ψ with V_B replaced by $V^{(M)}$.

Step 2 (clustering for the bounded part). $V^{(M)} \in \mathfrak{A}(B)$ and $A \in \mathfrak{A}(O)$ with $\text{dist}(O, B) = R$; the argument of [3, Thm. 2.3] applied at separation $R-|t|$ gives $|\Psi^{(M)}(t)| \leq 2M\|A\| e^{-\gamma(R-|t|)/2}$.

Step 3 (optimise). Choose $M := \exp(\frac{\gamma(R-|t|)}{2(1+\delta)})$. Then $M e^{-\gamma(R-|t|)/2} = M^{-\delta} = e^{-\gamma'(R-|t|)}$, and adding the two bounds gives the claim. $\square$

Remark 3.3 (a first moment is *not* enough). (F_1) supplies $\sup_g \mathbb{E}_{\mu_g}[|V_B|] < \infty$ uniformly. That alone does not bound $\mathbb{E}_{\mu_g}[|V_B| \mathbf{1}_{\{|V_B| > M\}}]$ uniformly in g : what is required is *uniform integrability* of the family $\{V_B\}$ under $\{\mu_g\}$, and (H6') is precisely the standard sufficient condition for it. So the gap between what is known and what is needed is exactly “ L^1 uniformly” versus “ $L^{1+\delta}$ uniformly”.

Lemma 3.4 (the far-time regime needs nothing new). *For all $t \in \mathbb{R}$, $|\Psi(t)| \leq K(A) := \sqrt{c}(1+\varepsilon)(\|(H(g)+c)^{1/2} A \Omega_g\| + \sqrt{c}\|A\|)$, uniformly in g and t .*

Proof. $\Psi(t) = \langle \Omega_g, V_B(\alpha_t(A) - \omega_g(A))\Omega_g \rangle$; insert $(H + c)^{\pm 1/2}$ on both sides of V_B and use (F₁), $\|(H + c)^{1/2}\Omega_g\| = \sqrt{c}$, and $\|(H + c)^{1/2}\alpha_t(A)\Omega_g\| = \|(H + c)^{1/2}e^{itH}A\Omega_g\| = \|(H + c)^{1/2}A\Omega_g\|$, which is t -independent and uniformly bounded by (F₂). $\square$

Theorem 3.2 and Lemma 3.4 together supply exactly the two regimes the M4 estimate needs, with (H6) replaced by (H6') and no other new input.

4 What (H6') still requires

Proposition 4.1. *Let μ_g^B, φ_0^B denote the restrictions to the σ -algebra generated by the field in B . If for some $q > 1$*

$$\sup_{g \in \mathcal{G}} \left\| \frac{d\mu_g^B}{d\varphi_0^B} \right\|_{L^q(\varphi_0^B)} < \infty, \quad (3)$$

then (H6') holds for every $\delta > 0$; and if (3) holds for some $q > 2$ then (H6) holds.

Proof. V_B is $\mathfrak{A}(B)$ -measurable, so $\mathbb{E}_{\mu_g}[|V_B|^r] = \mathbb{E}_{\mu_g^B}[|V_B|^r]$. By Hölder with $1/p + 1/q = 1$, $\mathbb{E}_{\mu_g^B}[|V_B|^r] \leq \| |V_B|^r \|_{L^p(\varphi_0^B)} \|d\mu_g^B/d\varphi_0^B\|_{L^q}$, and $\| |V_B|^r \|_{L^p(\varphi_0)} = \|V_B\|_{L^{rp}(\varphi_0)}^r < \infty$ for every $rp < \infty$ by Nelson's hypercontractive bound, with a constant depending only on B, P and not on g . $\square$

Remark 4.2 (this is GRS Problem 1, verbatim). (3) is Problem 1 of [7, p. 125]: “prove that for some fixed $p > 1$, $\|\nu_{\Lambda\Lambda'}\|_p$ is bounded independently of Λ' ”, with the conjecture $\log \|\nu_{\Lambda\Lambda'}\|_p = O(|\Lambda|)$. GRS observe that such a bound would re-prove the locally Fock property and give the infinite-volume equal-time vacuum expectation values. So (H6') is not a new problem; it is a fifty-year-old one, and Theorem 3.2 shows the weakest form of it ($q > 1$, arbitrarily close to 1) already suffices for M4.

Remark 4.3 (the N_τ route, for comparison). The localised N_τ estimates of [4, (3.2.9), (4.19)] give $\|V_B\Omega_g\| \leq \text{const} \cdot \|w_B\| \cdot \|(N_{B'} + I)^{n/2}\Omega_g\|$ for a degree- n Wick monomial, so (H6) follows from $\sup_g \omega_g(N_{B'}^n) < \infty$ with $n = \deg P$ — the $\deg P$ -th moment of the local number operator, strictly stronger than (H2)-full's first moment. This route is therefore worse than Proposition 4.1.

5 Status and what I would try next

- **Proved:** the gap-based routes are vacuous, for an ultraviolet reason (Props. 2.1–2.2, Rem. 2.4); (H6) may be replaced by the strictly weaker (H6') at a rate cost $\delta/(1 + \delta)$ (Thm. 3.2); the far-time regime needs no new input (Lem. 3.4).
- **Not proved:** (H6'). It is implied by, and appears to need, a uniform local L^q bound on the ground-state density with $q > 1$ — GRS Problem 1.
- **Most promising input.** Bauerschmidt–Dagallier's log-Sobolev inequality for the φ_2^4 measure is uniform in the volume *and* in the lattice regularisation. LSI is stable under marginalisation, so a uniform LSI for the space-cutoff Euclidean measure ν_g descends to its time-zero marginal μ_g ; and an LSI yields Gaussian concentration for suitable functionals, which is the natural source of uniform $L^{1+\delta}$ bounds on a fixed local V_B . Whether this can be pushed to (3), or directly to (H6') for Wick polynomials rather than Lipschitz functions, is the concrete question. It is already on the M5 critical path, so the cost of checking is shared.

6 Two candidate routes to (H6'), and a pattern

Route 1: Gaussian domination of moments — blocked exactly at Wick powers

For even P in the Griffiths–Simon class the lattice approximation is a genuine ferromagnet [7, §IV.4], and there the machinery exists: Sylvester [8] and Shlosman [9] give the signs of the Ursell functions ($u_{2n} \leq 0$ for $n \geq 2$), and Newman [10], Ellis–Monroe–Newman [11] convert these into *Gaussian upper bounds*

$$\langle \varphi_{x_1} \cdots \varphi_{x_{2n}} \rangle \leq \sum_{\text{pairings}} \prod \langle \varphi \varphi \rangle.$$

If such a bound survived to the continuum, (H6) would follow *outright*, and with the *free-field* value as the uniform constant: $\mathbb{E}_{\mu_g}[V_B^2] \leq \mathbb{E}_{\varphi_0}[V_B^2] = (\deg P)! \iint h h C^{\deg P} < \infty$, finite because $C(x - y) \sim -\frac{1}{2\pi} \log |x - y|$ is locally L^p for every p in one dimension.

The obstruction is the Wick subtraction, and it is documented. On the lattice there is no Wick ordering; in the continuum limit the counterterms subtract $C_\varepsilon(0) \sim \log(1/\varepsilon)$, and an *inequality* does not survive subtracting a divergent quantity from both sides. This is exactly the complaint of [7, §V.3]: “the fact that the interaction involves Wick powers limits the applicability of the correlation inequalities ... we are able to handle only a single quadratic power $:\varphi^2:$ ”; and the situation has not materially changed.

The concrete sub-problem, therefore: does the Gaussian bound survive Wick subtraction, given $\langle \varphi(x)^2 \rangle_g \leq C(0)$ (expected for repulsive P)? This is a finite combinatorial question — matching counterterms on the two sides of the inequality — not an analytic one, and is worth an afternoon.

Route 2: chessboard estimates — and a synergy with M1

Multiple-reflection (chessboard) estimates in the sense of Fröhlich–Israel–Lieb–Simon convert a *local* expectation into a universal bound controlled by the *pressure*, and the pressure is uniformly bounded here: $-MV \leq E_g \leq 0$ [4, Thm. 5.1] and $\alpha(l) \uparrow \alpha_\infty < \infty$ [6, Thm. 1]. This route does not care about Wick powers in the way Route 1 does.

Tension: reflection positivity requires the cutoff to respect the reflection hyperplanes, which a generic g does not. *Resolution*: by [2, Thm. 3.3] we are free to choose the cofinal family — the Chebyshev selection already exploits that freedom — so one may take reflection-symmetric or periodic cutoffs. This is a genuine opening and costs nothing structurally.

Dead ends, recorded so they are not repeated

1. *Interpolating from the first moment.* $\mathbb{E}[|V|^{1+\delta}] \leq \mathbb{E}[|V|]^{1-\theta} \mathbb{E}[|V|^r]^\theta$ needs a higher moment: circular.
2. [6, Thm. 2] with $Q = \pm \varepsilon P^2$. $P + \varepsilon P^2$ is semibounded, so one obtains only $\omega_g(\int h: P^2(\varphi):) \geq -c/\varepsilon$ — a *lower* bound. $P - \varepsilon P^2$ is not semibounded, so the theorem gives no upper bound. (And $\int h: P^2: \neq V_B^2$ in any case.)
3. *Euclidean conditioning.* Writing the local density via the Markov property produces an uncontrolled boundary functional — that *is* Problem 1.
4. *The spectral gap.* Dead, for the ultraviolet reason of §2.

A pattern worth noticing

Correlation inequalities are available for $:\varphi^2:$ and no further. That is the same boundary as M2's solvable model (quadratic P), and the same boundary as the FKG/Daubechies dichotomy of the wavelet programme (ferromagnetic structure at $K = 1$, lost for $K \geq 2$). Three independent places where the difficulty is exactly “*positivity is available at the bottom of the hierarchy and lost above*”. This is evidence that the log-Sobolev / Polchinski route — which does not require positivity — is the structurally correct bet, and it is already the M5 plan.

References

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